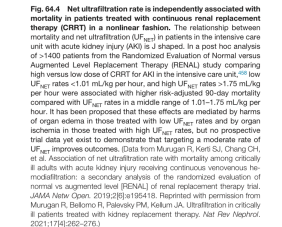
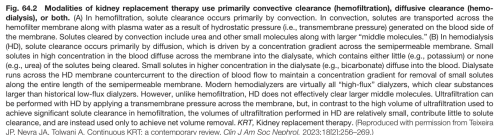


Fig_64_1



Fig_64_4



Fig_64_2

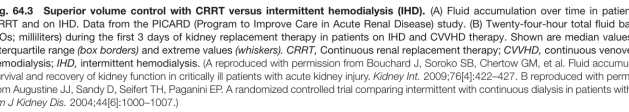


Fig. 64.3 Superior volume control with CRRT versus intermittent hemodialysis (IHD). (A) Fluid accumulation over time in patient on CRRT and on IHD. Data from the PICARD Program to Improve Care in Acute Renal Disease (PICARD). (B) Twenty-four-hour total fluid balance in patients on CRRT and on IHD. Data from the PICARD Program to Improve Care in Acute Renal Disease (PICARD). (C) Mortality during the first 3 days of kidney replacement therapy in patients on IHD and CVVHD therapy. Shown are median values (interquartile range [box borders] and extreme values [whiskers]). CRRT, Continuous renal replacement therapy; CVVHD, continuous veno-venous hemodialysis; IHD, intermittent hemodialysis. (A reproduced with permission from Bouchard J, Soroko SB, Chertow GM, et al. Fluid accumulation during continuous renal replacement therapy in patients with acute renal failure. *Crit Care Med* 2000;28:1000-1007. B reproduced with permission from Augustine JN, Sandy D, Siefert TH, Paganini EP. A randomized controlled trial comparing intermittent with continuous dialysis in patients with acute renal failure. *N Engl J Med* 2004;351:1000-1007.)

Fig_64_3

Etiologies/Contributing Factors^a	Risk Factors/Predisposing Conditions^b
- Sepsis/septic shock: 40%–50% - Major surgery: 35% - Volume depletion: 25%–35% - Cardiogenic shock: 15%–25% - Drug toxicity: 15%–20% - Hepatorenal syndrome: 3%–6% - Urinary tract obstruction: 1%–3%	- Chronic kidney disease - Prior AKI - Advanced age - Frailty¹¹⁸ - Nonrenal comorbidities: - Diabetes - Heart disease, especially CHF - Liver disease - Malignancy - Treatment with diuretics, NSAIDs, or RAAS inhibitors

^aThe frequencies of these etiologies/factors are derived from Uchino S, Kellum JA, Bellomo R, et al. *A multicenter international study of the frequency of patients: a multinational, multicenter study*. *JAMA*. 2005;294(7):813–818 and Hoste EA, Bagshaw SM, Bellomo R, et al. Epidemiology of acute kidney injury in critically ill patients: the multinational AKI-EPI study. *Intensive Care Med*. 2015;48(11):1411–1423 but are rounded for simplicity.

^bCHF: Congestive heart failure; NSAID, nonsteroidal anti-inflammatory drug; RAAS, renin-angiotensin-aldosterone system.

Table_64_1

[illegible]

Table_64_2

[illegible]

Table_64_3

Complication	Mechanism	Diagnostic Features	Management Options
Excess buffer from citrate	Excess citrate supply, resulting in excessive metabolic conversion of citrate to bicarbonate and excess buffer	• Metabolic alkalosis • $\text{rCa}/\text{Ca} < 2.5$	• Decrease blood flow rate/citrate rate^a • Increase dialysate rate • Decrease buffer concentration in other CRRT solutions
Citrate accumulation ^b	Impaired metabolic conversion of citrate resulting in accumulation of citrate-calcium complexes in blood	• Anion gap metabolic acidosis • $\text{rCa}/\text{Ca} < 2.5$ • Escalating calcium infusion requirements	• Decrease blood flow rate/citrate rate^a • Increase dialysate rate • Discontinue regional citrate anticoagulation
Insufficient buffer from citrate	Inadequate citrate supply, resulting in insufficient conversion of citrate to bicarbonate and insufficient buffer	• Metabolic acidosis • $\text{rCa}/\text{Ca} < 2.5$	• Increase blood flow rate/citrate rate^a • Decrease dialysate rate • Increase buffer concentration in other CRRT solutions

^aThese changes in blood flow and citrate rate are done proactively, to increase or decrease the delivery of citrate while maintaining the same ratio of citrate infusion rate to blood flow rate and thus similar inflator/inlet calcium concentration.

^bCitrate accumulation is also known as citrate toxicity or citrate lock.

CRRT, Continuous renal replacement therapy; Ca, ionized calcium; rCa, total calcium.

Adapted with permission from Neyra JA, Yessayan L, Thompson Bastin ML, et al. How to prescribe and troubleshoot continuous renal replacement therapy: a case-based review. *Kidney360*. 2021;2(2):371–384.

Table_64_4

Table 64.2 Factors to Consider When Deliberating Timing of Kidney Replacement Therapy (KRT) Initiation for Acute Kidney Injury (AKI) Patients in the Intensive Care Unit (ICU)

Illness Severity and Trajectory

- AKI severity and trend
- Fluid balance and symptoms of fluid overload
- Severity of electrolyte and acid-base disorders
- Presence of relevant nonrenal organ dysfunction impacted by AKI and/or fluid overload
- Anticipated need for enhanced metabolic support (e.g., rhabdomyolysis or tumor lysis syndrome)

Necessity of KRT

- Presence of oliguria, considering the effect of diuretics
- Likelihood of recovery of kidney function without KRT, considering the reversibility of the specific etiology of AKI, trends in kidney function and urine output, and baseline kidney function
- Underlying comorbidities impacted by AKI and/or fluid overload

Risks of KRT

- Hemodynamic instability from KRT
- Infection, including catheter-related bloodstream infection
- Other risks associated with vascular access (pneumothorax, procedural bleeding, catheter-associated deep venous thrombosis, etc.)
- Clearance of trace elements, water-soluble vitamins, phosphate, amino acids/small peptides, and drugs (especially antibiotics)
- Delayed renal recovery
- Increased risk of immobilization / impeding physical rehabilitation

Other Factors

- Patient and family wishes and overall goals of care
- Overall prognosis including likelihood of patient survival
- Availability of machines, disposable supplies, and nursing staff
- Health care costs

Adapted from Macedo E, Mehta RL. Continuous dialysis therapies: core curriculum 2016. Am J Kidney Dis. 2016;68(4):645–657 and Ostermann M, Joannidis M, Puri A, et al. Patient selection and timing of continuous renal replacement therapy. *Blood Purif*. 2016;42(2):224–237.

Table_e64_2